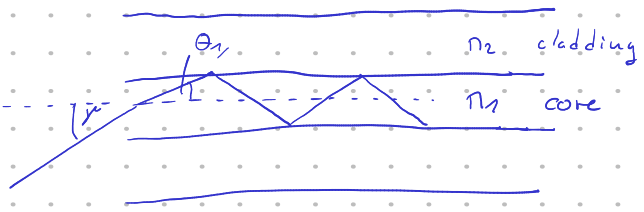
AOTC Exercise 04

11.11.2025



$\theta_1 < \theta_g$ θ_1 has to be smaller than θ_g to achieve TIR

$$\sin\theta_{1g} = \sqrt{1-\cos^2\theta_{1g}} = \frac{1}{n_1}\sqrt{n_1^2-n_2^2} \quad (1)$$

$$\sin\gamma = n_1 \cdot \sin\theta_{1g} = \sqrt{n_1^2-n_2^2} = A_N$$

A_N => numerical Aperture of the Fiber

Field inside the stepinder fiber

$$\vec{E}(x,y,z) = \vec{E}_0(x,y)\, e^{-i\beta z} \quad (3) \qquad \beta = k_0 \cdot n_1 \cdot \cos(\theta_1)$$

with $\theta_1 < \theta_{1g}$ & $k_0 n_2 < \beta < k_0 n_1$

Helmholtz equation $\Delta \underline{E}_{x,y} + k_0^2 n_i^2 \underline{E}_{x,y} = 0$ & $i = 1,2$ for cladding and core

$$-\beta^2 \cdot \vec{\underline{E}}_{x,y} = \frac{\partial^2 \underline{E}_{x,y}}{\partial z^2}$$ field will only propagate in z, direction

$$\Delta_t \underline{E}_{x,y} + \frac{\partial^2}{\partial z^2} E_{xy} + k_0^2 n_i^2 \underline{E}_{x,y} = 0 \quad \& \quad \Delta_t = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$ only transversal

$$\Delta_t \underline{E}_{x,y} + \underbrace{\left(k_0^2 \cdot n_i^2 - \beta^2\right)}\, \underline{E}_{x,y} = 0$$

for n_1 $k_0^2 n_1^2 - \beta^2 > 0$ -> field will oscillate inside the core

for n_2 $k_0^2 n_2^2 - \beta^2 < 0$ -> field decay in the cladding

for Standard fiber, low index contrast, only weakly guiding approximation

$$\frac{n_1 - n_2}{n_1} \ll 1$$

&

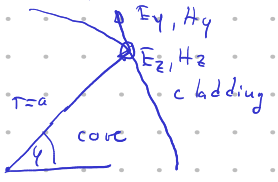
field envelope, only varies slowly in z-direction, „slowly varying envelope approximation"!

$$\left|\frac{\partial}{\partial x}\right|, \left|\frac{\partial}{\partial y}\right| \ll \beta$$ ← transverse derivatives only vary slowly, compared to β (transverse ~ 10µm, longitudinal ~ 1,5µm)

for step index fibers, weak guidance, and slow variation of the field envelope in z-direction are assumed.

for field distributions inside the fiber, we will consider only $\underline{E}_x$, and $\underline{E}_y$ components.

for a centro-Symmetric fiber we have change to clyndric coordinates:



r=a E_z, H_z & E_y, H_y are assumed to be steady

from Maxwell eq. $-\nabla \times \vec{E} = i\omega\mu \vec{H}$

$$H_z = -i\omega\mu_0 \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)$$

another Maxwell Eq. $\nabla \times H = i\omega \varepsilon_0 n_i^2 \vec{E}$

$$\underline{E}_x = \frac{1}{i\omega\varepsilon_0 n_i^2}\left(\frac{\partial \underline{H}_z}{\partial y} + i\beta \underline{H}_y\right)$$

$$\underline{E}_y = \frac{1}{i\omega\varepsilon_0 n_i^2}\left(-\frac{\partial \underline{H}_z}{\partial x} - i\beta \underline{H}_x\right)$$

now we have $\underline{H}_x$, $\underline{H}_y$, and $\underline{H}_z$, all from E_x, E_y — this will fall out!

$$\underline{H}_x \hat{=} -\frac{\omega\varepsilon_0 n_i^2}{\beta}\cdot \underline{E}_y$$

$$\underline{H}_y \hat{=} \frac{\omega\varepsilon_0 n_i^2}{\beta}\underline{E}_x$$

Due to the slowly varying envelope approximation $\frac{\partial H_z}{\partial y} \ll \frac{\partial H_z}{\partial x}$ can be neglected

for $\underline{E}_z$

$$\underline{E}_z = \frac{1}{i\omega\varepsilon_0 n_i^2}\left(\frac{\partial \underline{H}_y}{\partial x} - \frac{\partial \underline{H}_x}{\partial y}\right) \approx \frac{1}{i\beta}\left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right)$$

now all 6 field components are given!

for cylindrical coordinates we need $\underline{E}_\varphi$, $\underline{E}_z$, $\underline{H}_\varphi$, $\underline{H}_z$

$$E_\varphi = E_y \cdot \cos\varphi - E_x \cdot \sin\varphi$$

$$H_\varphi = H_y \cdot \cos\varphi - H_x \cdot \sin\varphi$$

$$E_z = 1/i\beta\left(\cos\varphi \cdot \frac{\partial E_x}{\partial r} - \frac{1}{r}\cdot \sin\varphi \frac{\partial E_x}{\partial \varphi} + \sin\varphi \frac{\partial E_y}{\partial r} + \frac{1}{r}\cdot\cos\varphi \frac{\partial E_y}{\partial \varphi}\right)$$

$$H_z = 1/i\omega\mu_0\left(\cos\varphi \frac{\partial E_y}{\partial r} - \frac{1}{r}\cdot\sin\varphi\frac{\partial E_y}{\partial\varphi} + \sin\varphi\frac{\partial E_x}{\partial r} - \frac{1}{r}\cdot\cos\varphi\frac{\partial E_x}{\partial\varphi}\right)$$

now we have all necessary cylindric field components.

for $r=a$ E_x and E_y aren't coupling with each other, there should be waves which are solutions to the Helmholtz equation and be linear polarized.

$$\underline{E}_x = \psi(r,\varphi)\cdot e^{-i\beta z} \qquad \psi(r,\varphi) = \begin{cases} \psi_1(r,\varphi) & \text{for } r \le a \\ \psi_2(r,\varphi) & \text{for } r > a \end{cases}$$

now we can rewrite the Helmholtz equation as a scalar equation:

$$\Delta_t \psi_i + \left(k_0^2 n_i^2 - \beta^2\right)\cdot\psi_i = 0 \qquad i = 1,2$$

$$\psi_1\big|_{r=a} = \psi\big|_{r=a}$$

$$\frac{\partial\psi_1}{\partial r}\bigg|_{r=a} = \frac{\partial\psi_2}{\partial r}\bigg|_{r=a}$$

no jumps in field at the core–ladding boundry

fiber parameter $V = k_0 \cdot a\sqrt{n_1^2 - n_2^2} = k_0 \cdot a \cdot A_N$

normalized propagation constant $B = \frac{\beta^2/k_0^2 - n_2^2}{n_1^2 - n_2^2} = \frac{\beta/k_0 - n_2}{(n_1 - n_2)} \quad \frac{\beta/k_0 - n_2}{(n_1 - n_2)} \approx \frac{\beta/k_0 - n_2}{n_1 - n_2}$

core parameter $u = V\sqrt{1-B} = a\cdot\sqrt{k_0^2 n_1^2 - \beta^2}$

cladding parameter $v = V\sqrt{B} = a\cdot\sqrt{\beta^2 - k_0^2 n_2^2}$

rewriting the Helmholtz-equation with the normalizations:

$$a^2 \Delta_t \Psi_1 + U^2 \Psi_1 = 0 \qquad \text{for } r \leq a$$

$$a^2 \Delta_t \Psi_2 + V \Psi_2 = 0 \qquad \text{for } r > a$$

solving those differential equations:

$$\Psi_1(r,\varphi) = A_1 \cdot J_\ell\left(r\frac{U}{a}\right) \begin{Bmatrix} \cos(\ell\cdot\varphi) \\ \sin(\ell\cdot\varphi) \end{Bmatrix}$$

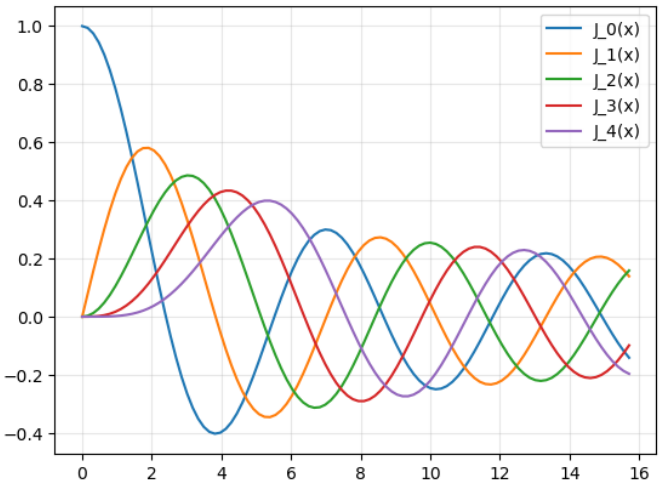
↑ Bessel function of first kind

inside the core, with an oscillating field!

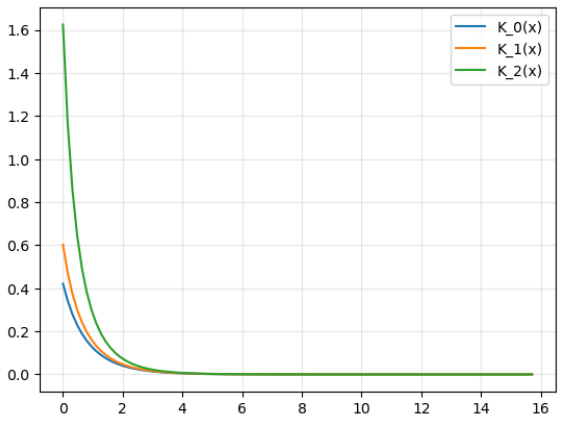
$$\Psi_2(r,\varphi) = A_1 \, K_\ell\left(r\frac{U}{a}\right) \begin{Bmatrix} \cos(\ell\cdot\varphi) \\ \sin(\ell\cdot\varphi) \end{Bmatrix}$$

↑ modified Hankel function

ℓ is an integer 1, 2, 3 ....



Bessel functions, (core)



Hankel functions (cladding)

If one divides both equations, we get a characteristic equation

$$-\frac{U \cdot J_{\ell+1}(U)}{J_\ell(U)} + V \cdot \frac{K_{\ell+1}(V)}{K_\ell(V)} = 0$$

U and V are only dependent on B

$$U = V\sqrt{1-B} \quad \& \quad V = V\sqrt{B}$$

so ℓ is the circumferencial order 0, 1, 2, 3 4

when departing from 0 (ℓ), there will more then one solution for ℓ

Those solutions will be then denoted p.

Problem 9 a) $V = k_0 \cdot a \cdot \sqrt{n_1^2 - n_2^2} = \frac{2\pi}{1{,}55\,\mu m} \cdot 6{,}1\,\mu m \sqrt{1.457^2 - 1.444^2} = 4.802$ → plug this value into the B-V-Diagram

for modes are supported!

$B_{01} = 0.83$; $B_{11} = 0.58$; $B_{21} = 0.26$ $B_{02} = 0.17$